Calibration results
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Normalized Residuals

Reprojection error (cam0): mean 0.263029158727, median 0.239249090992, std: 0.150206808913
Reprojection error (cam1): mean 0.264175527025, median 0.241648330808, std: 0.147940374662
Gyroscope error (imu0): mean 0.0988102021001, median 0.0786368870588, std: 0.126567488981
Accelerometer error (imu0): mean 0.158329251395, median 0.123598566412, std: 0.219979896301

Residuals

Reprojection error (cam0) [px]: mean 0.263029158727, median 0.239249090992, std: 0.150206808913
Reprojection error (cam1) [px]: mean 0.264175527025, median 0.241648330808, std: 0.147940374662
Gyroscope error (imu0) [rad/s]: mean 0.00312465294698, median 0.00248671671211, std: 0.0040024154
Accelerometer error (imu0) [m/s^2]: mean 0.0500681054636, median 0.0390852985395, std: 0.069563751

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.99580657 -0.09130022 -0.00579234 0.04673254]
 [-0.00480648 0.11544191 -0.9933026 -0.00542932]
 [0.09135743 -0.98910942 -0.11539664 -0.02601016]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.99580657 -0.00480648 0.09135743 0.0488867]
 [-0.09130022 0.11544191 -0.98910942 -0.02083343]
 [-0.00579234 -0.9933026 -0.11539664 -0.00812375]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.00191028630033

Transformation (cam1):

T_ci: (imu0 to cam1):

```
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[-0.99762178  0.00108522 -0.06891744 -0.08774245]
 [ 0.0683667   0.14269696 -0.98740244 -0.02293287]
 [ 0.00876276 -0.98976583 -0.14243179 -0.00796936]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.00174981612969

Baselines:
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Baseline (cam0 to cam1):
[[ 0.98714566  0.00398336 -0.1597735  -0.13616204]
 [-0.00837587  0.99960497 -0.02682811  0.00060066]
 [ 0.15960352  0.0278215   0.98678908 -0.01146713]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.136645364441 [m]


Gravity vector in target coords: [m/s^2]
[ 5.56111468  0.07888884 -8.07689315]


Calibration configuration
=========================

cam0
-----
  Camera model: pinhole
  Focal length: [231.21241102886728, 230.12786976767822]
  Principal point: [241.16196202316743, 323.138612753848]
  Distortion model: equidistant
  Distortion coefficients: [0.03350156917979561, 0.041167994039106835, -0.05091722480035919, 0.01342
  Type: aprilgrid
  Tags:
```

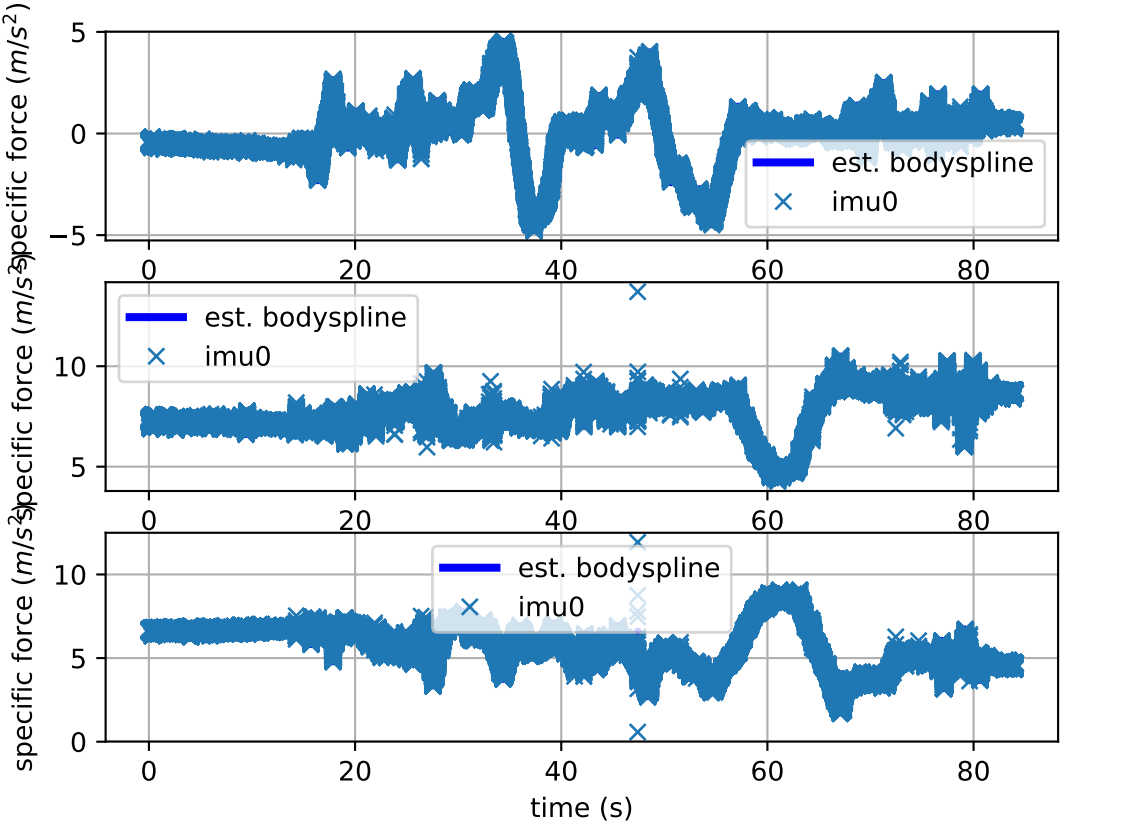
```
    Spacing 0.010372 [m]

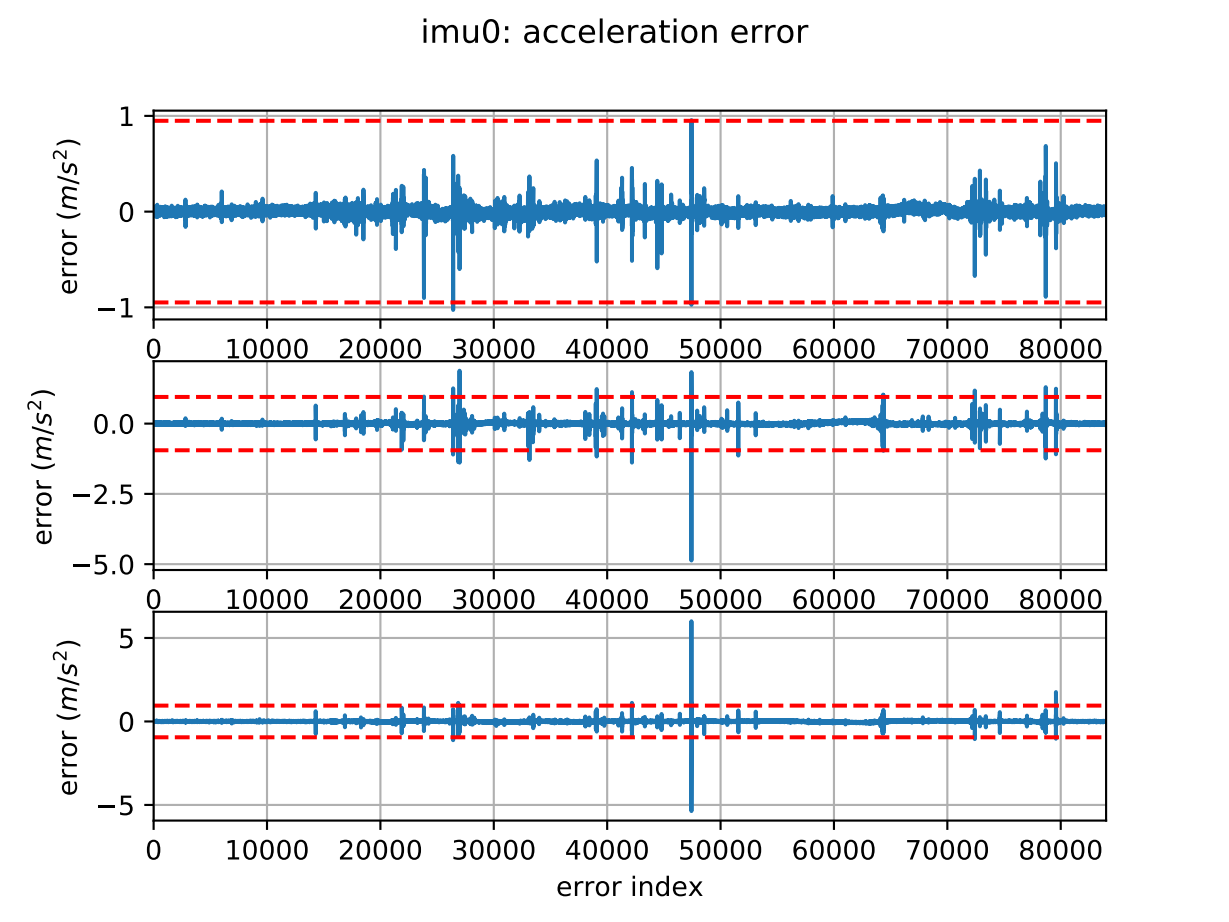

cam1
-----
  Camera model: pinhole
  Focal length: [230.33277762866882, 229.0276560626669]
  Principal point: [239.3694217273709, 314.239098804625]
  Distortion model: equidistant
  Distortion coefficients: [0.02606597634347255, 0.04780113790413336, -0.03883241654779152, 0.006895
  Type: aprilgrid
  Tags:
    Rows: 7
    Cols: 5
    Size: 0.02593 [m]
    Spacing 0.010372 [m]

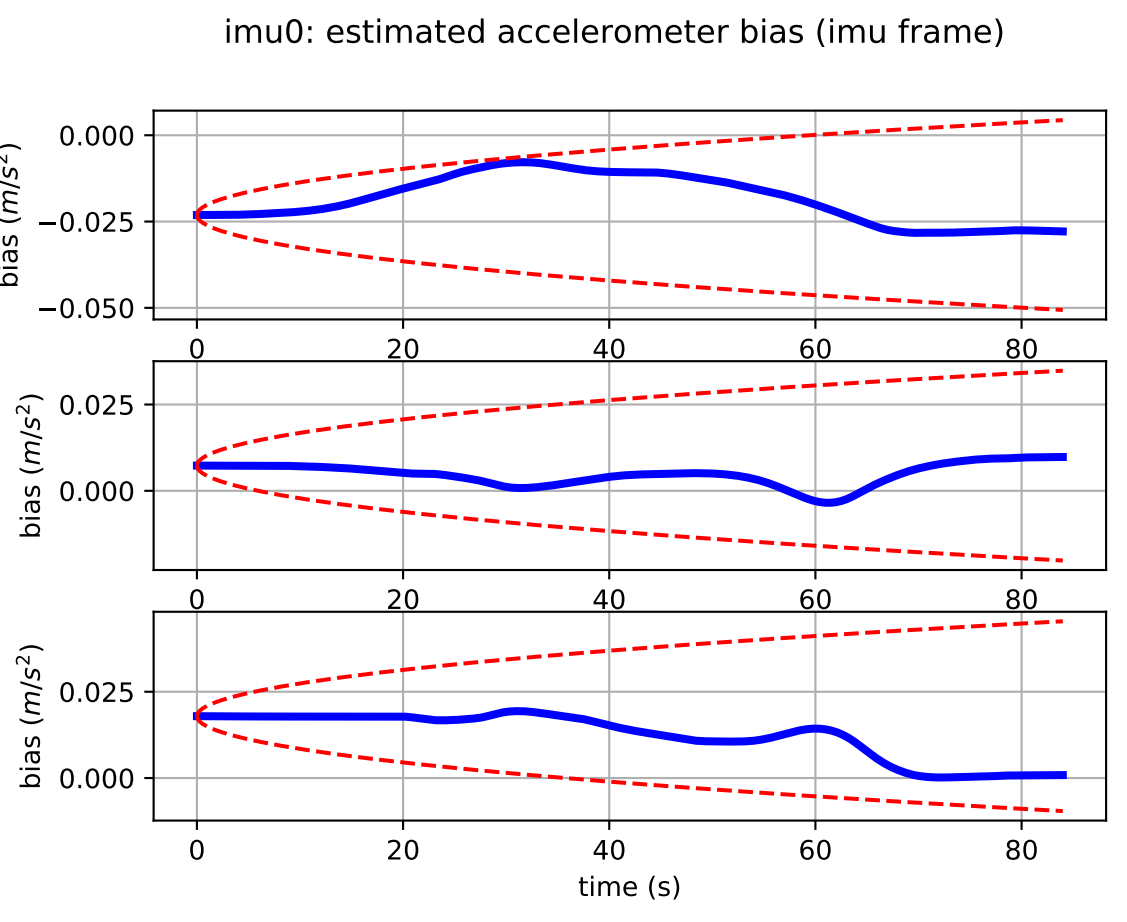


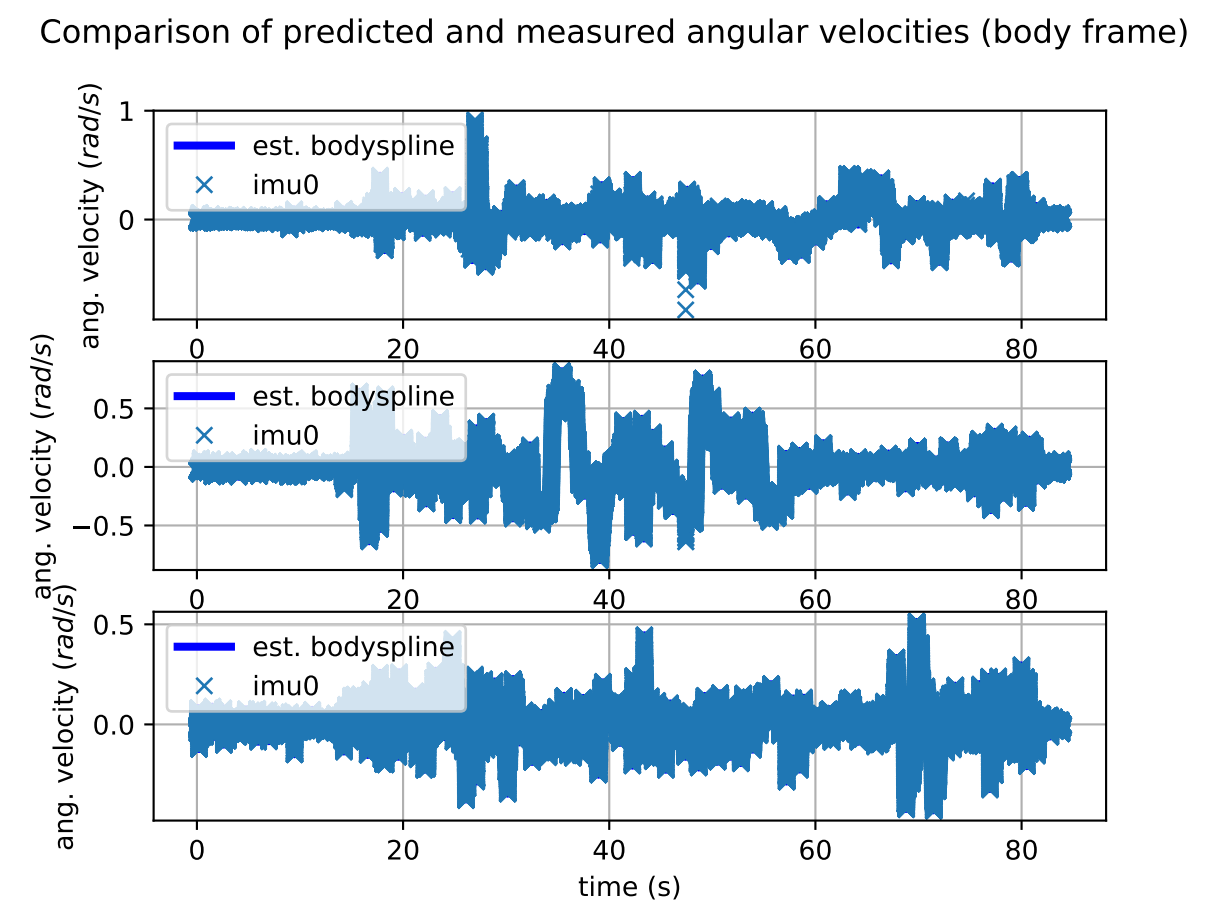
IMU configuration
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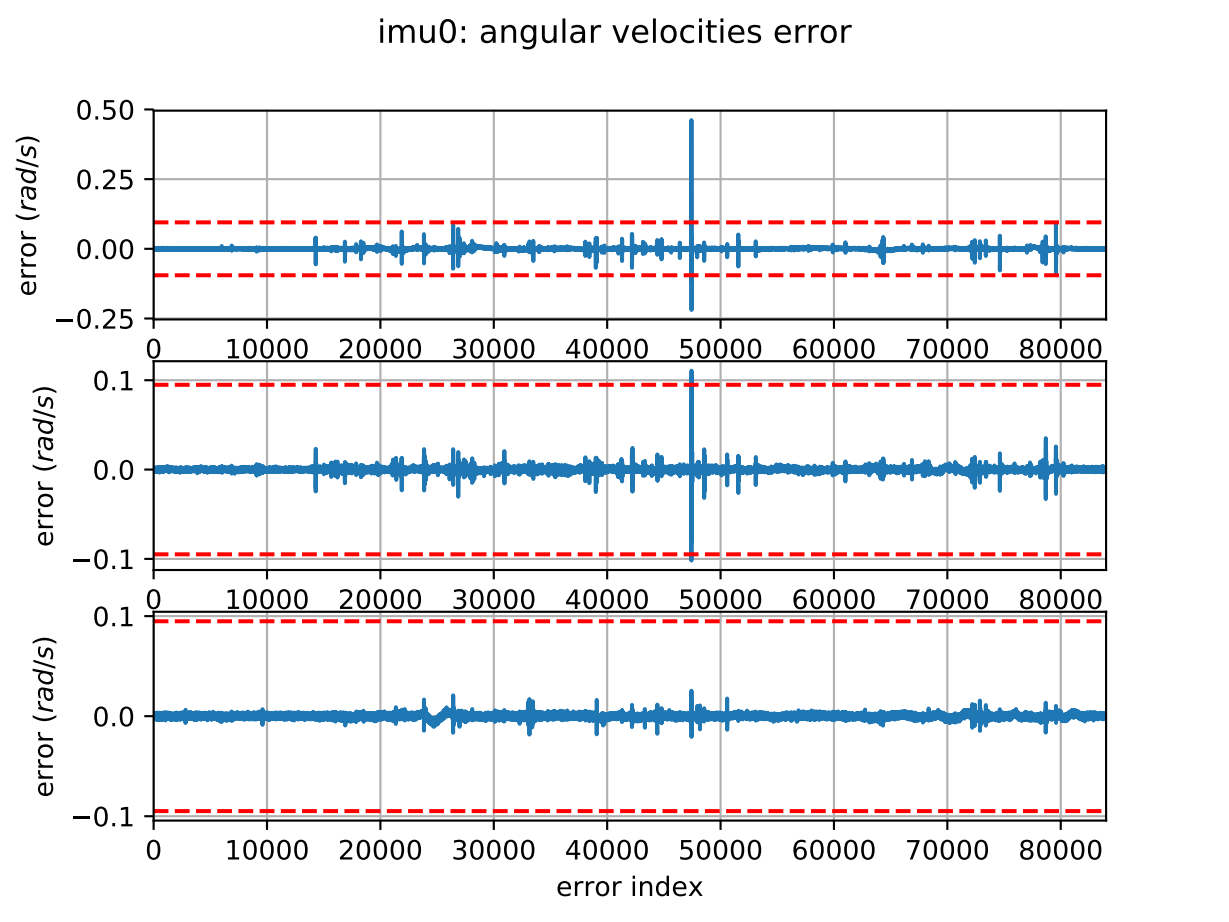
IMU0:
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  Model: calibrated
  Update rate: 1000.0
  Accelerometer:
    Noise density: 0.01
    Noise density (discrete): 0.316227766017
    Random walk: 0.001
  Gyroscope:
    Noise density: 0.001
    Noise density (discrete): 0.0316227766017
    Random walk: 0.0001
  T_i_b
    [[ 1.  0.  0.  0.]
     [ 0.  1.  0.  0.]
```

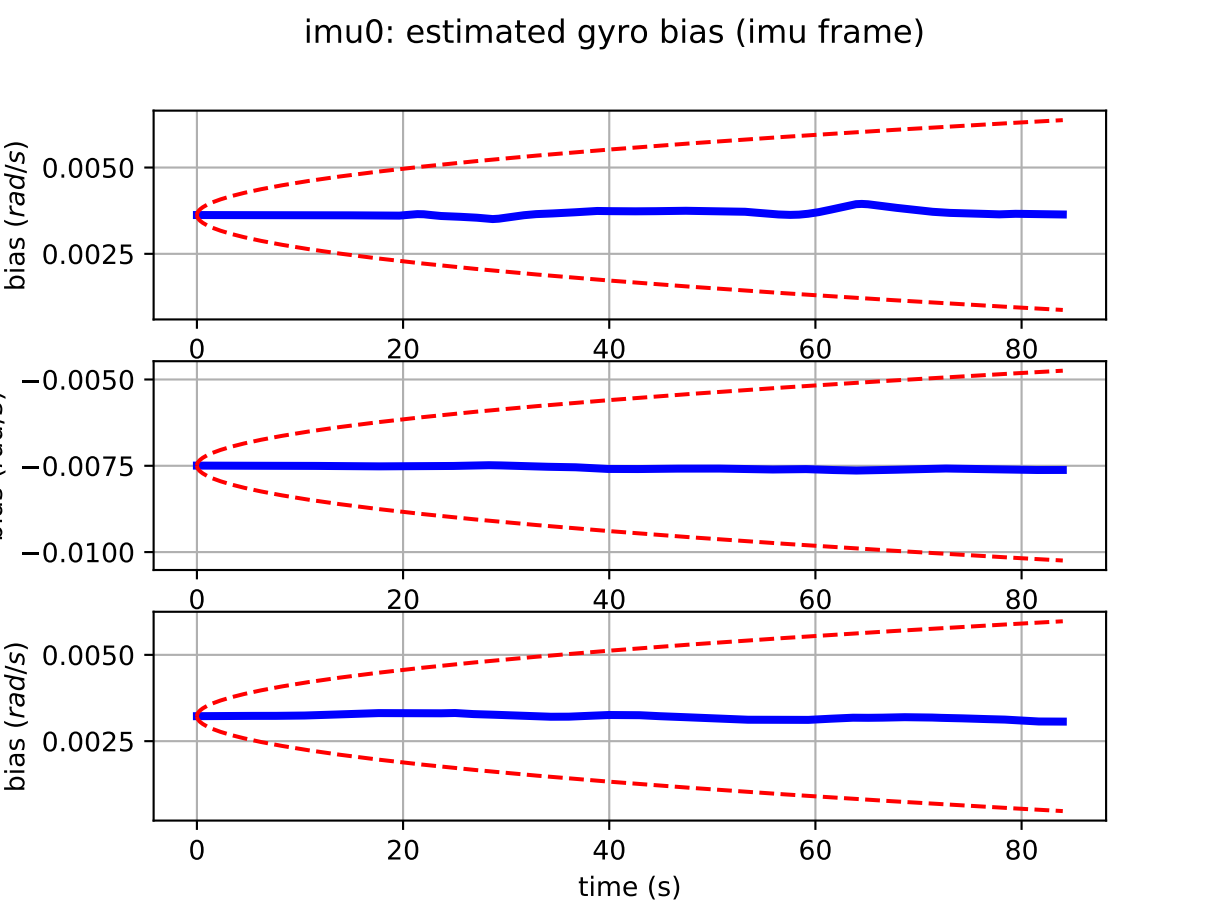

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)

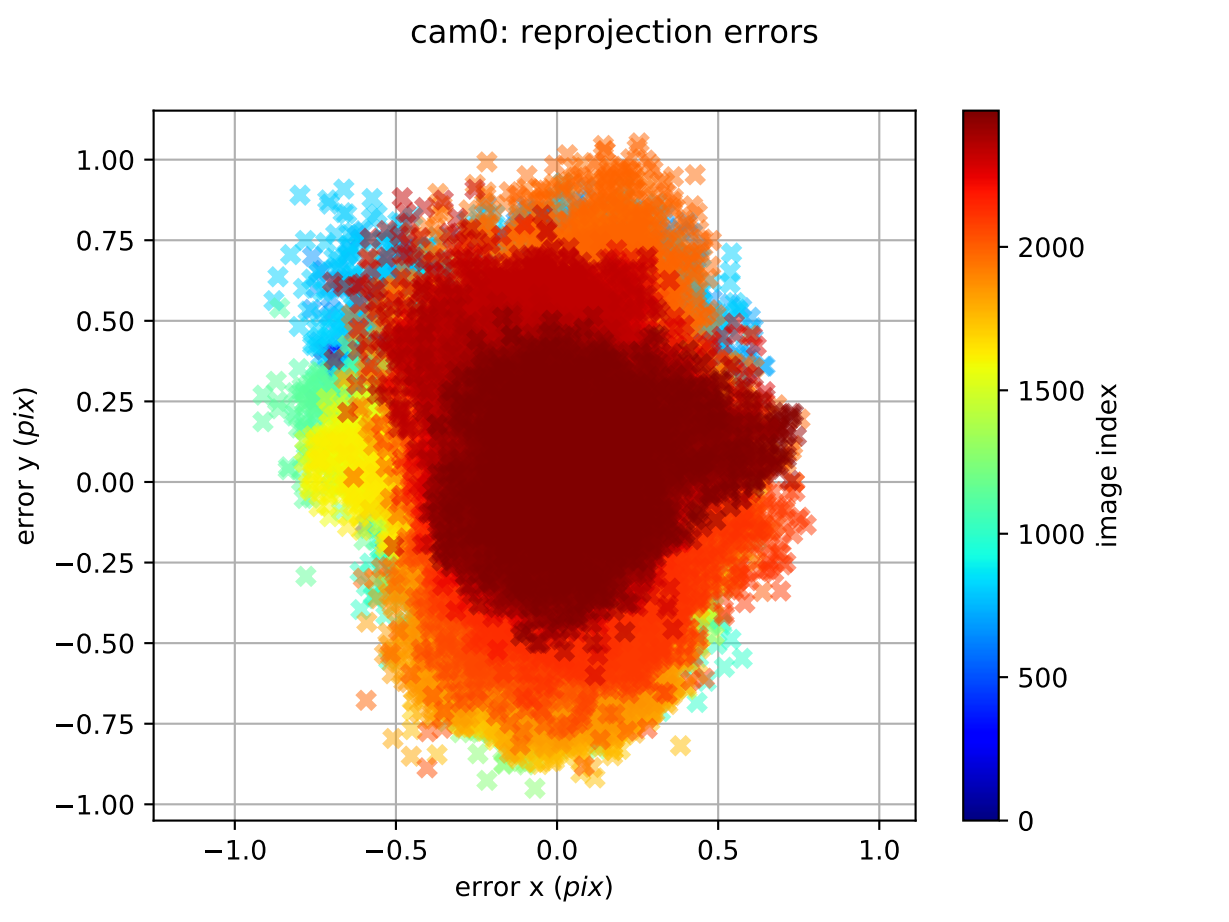

imu0: acceleration error
error (m/s^2)
1
0
−1
0
10000
20000
30000
40000
50000
60000
70000
80000
error (m/s^2)
0.0
−2.5
−5.0
0
10000
20000
30000
40000
50000
60000
70000
80000
error (m/s^2)
5
0
−5
0
10000
20000
30000
40000
50000
60000
70000
80000
error index

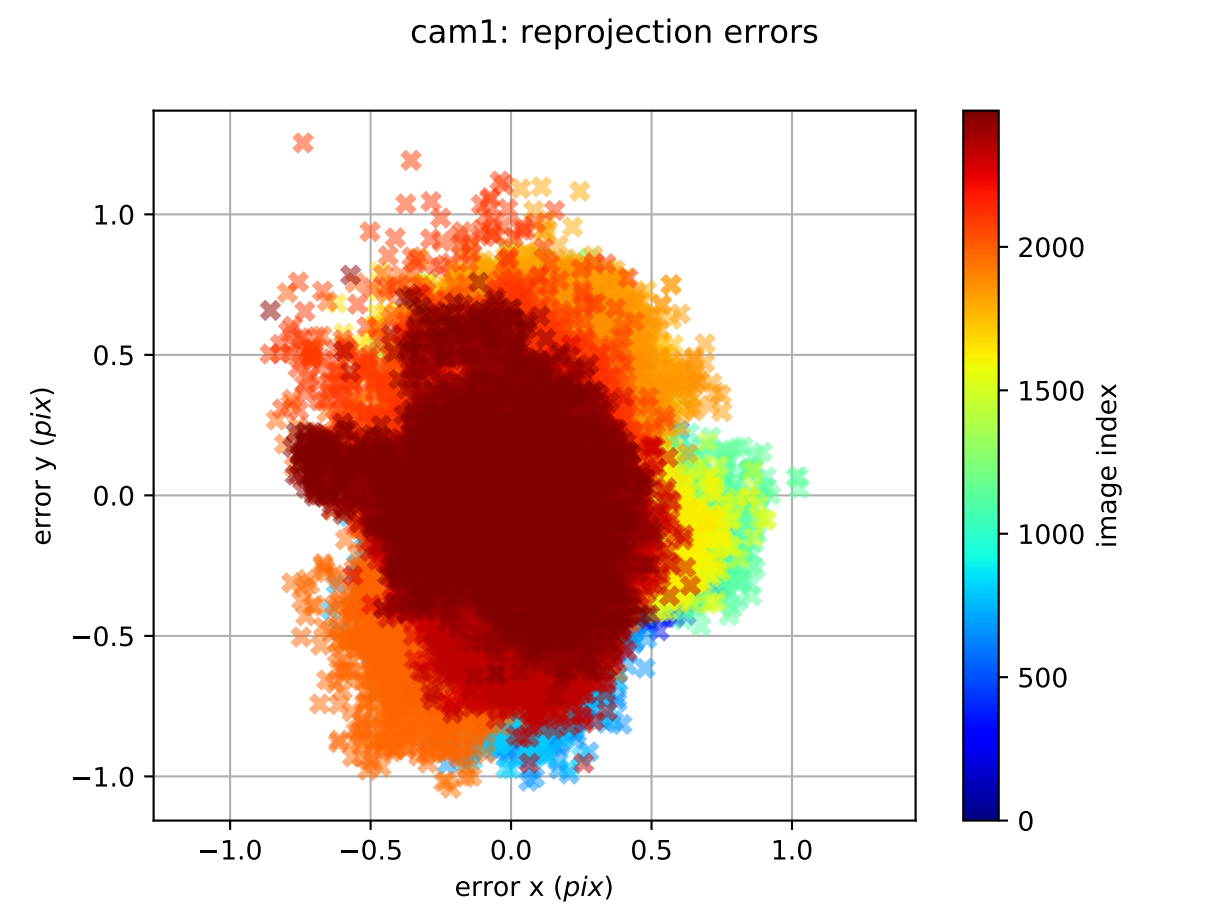

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.000
−0.025
−0.050
0
20
40
60
80
bias (m/s^2)
0.025
0.000
bias (m/s^2)
0.025
0.000
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
1
0
0
20
40
60
80
ang. velocity (rad/s)
est. bodyspline
imu0
0.5
0.0
−0.5
0
20
40
60
80
ang. velocity (rad/s)
est. bodyspline
imu0
0.5
0.0
0
20
40
60
80
time (s)


imu0: angular velocities error
error (rad/s)
0.50
0.25
0.00
−0.25
0
10000
20000
30000
40000
50000
60000
70000
80000
error (rad/s)
0.1
0.0
−0.1
error (rad/s)
0.1
0.0
−0.1
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.0050
0.0025
0
20
40
60
80
−0.0050
−0.0075
−0.0100
bias (rad/s)
0.0050
0.0025
0
20
40
60
80
time (s)


cam0: reprojection errors
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
error y (pix)
−1.0
−0.5
0.0
0.5
1.0
error x (pix)
2000
1500
1000
500
0
image index


cam1: reprojection errors
error x (pix)
error y (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
2000
1500
1000
500
0